

Backend Practice Project (Exam Style) Task Management API

Build a complete backend API for a Task Management System using modern backend architecture and authentication flow. This project is designed to test your understanding of Express, MongoDB, JWT authentication, middleware, MVC structure, and REST APIs.

Tech Stack Requirements

- Express
- MongoDB + Mongoose
- JWT
- bcrypt
- Middleware
- MVC Structure
- Environment Variables

Required Folder Structure

- src/config
- src/controllers
- src/middleware
- src/models
- src/routes
- src/utils (optional)
- src/app.js
- src/server.js

User Model Requirements

- Fields: name, email, password, role, createdAt
- Email must be unique
- Password must be hashed
- Default role = user
- Allowed roles: user, admin

Task Model Requirements

- Fields: title, description, status, priority, createdBy, createdAt
- Status enum: pending, in-progress, completed

- Priority enum: low, medium, high
- createdBy should reference User model

Authentication Features

- POST /api/auth/register
- Hash password before saving
- Prevent duplicate emails
- POST /api/auth/login
- Verify password using bcrypt.compare()
- Generate JWT token
- Return token after successful login

Middleware Requirements

- verifyToken middleware
- Read token from Authorization header
- Verify JWT token
- Attach decoded user to req.user
- verifyAdmin middleware
- Allow only admin users

Task CRUD Features

- POST /api/tasks - Protected route
- Save logged-in user ID in createdBy
- GET /api/tasks - Protected route
- Use populate('createdBy')
- GET /api/tasks/:id
- PATCH /api/tasks/:id
- Only creator OR admin can update
- DELETE /api/tasks/:id
- Only admin can delete tasks

Bonus Features (Optional)

- Filtering: /api/tasks?status=pending
- Pagination: /api/tasks?page=1&limit;=5
- Search: /api/tasks?search=design
- Sorting: /api/tasks?sort=createdAt

Project Rules

- Do NOT put everything in one file
- Do NOT store plain passwords
- Use proper routes/controllers/middleware separation
- Use async/await properly
- Use proper status codes

What This Project Tests

- MVC architecture
- Express routing
- Middleware chaining
- JWT authentication flow
- Password hashing
- MongoDB relationships
- Authorization logic
- Protected APIs
- REST API design

Submission Instructions: When you finish the project, prepare: 1. Folder structure 2. Important backend files 3. Explanation of authentication flow 4. Explanation of middleware flow 5. Explanation of authorization logic The project should be treated like a professional backend practical exam.